

QUANTUM SIMULATION

Observation of Shapiro steps in an ultracold atomic Josephson junction

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The current-voltage characteristic of a driven superconducting Josephson junction displays discrete steps. This phenomenon, called the Shapiro steps, forms today's voltage standard. In this work, we report the observation of Shapiro steps in a driven Josephson junction in a gas of ultracold atoms. We demonstrate that the steps exhibit universal features and provide insight into the microscopic dissipative dynamics that we directly observe in the experiment. We find that the steps are directly connected to phonon emission and nucleation of solitonic excitations, whose dynamics we follow in space and time. The experimental results are underpinned by extensive numerical simulations based on classical-field dynamics and may enable metrological and fundamental advances.

The Josephson effect is one of the most fundamental phenomena in quantum science and technology, featuring a dissipationless electrical current through a tunnel barrier between two superconductors by virtue of the phase difference of the traversing Cooper pairs' wave function. Above a critical value of the current, a finite voltage occurs across the junction as a result of the formation of superconducting quasiparticles (1). The Josephson effect and Josephson junctions have played prominent roles in understanding the notion of macroscopic quantum coherence, which led to important technological applications, such as superconducting quantum interference devices (SQUIDS) (2–6); they are also the core ingredients of superconducting qubits (7–10). If an additional microwave field is applied across the Josephson junction, a staircase current-voltage characteristic emerges, referred to as Shapiro steps (11). The origin of this effect is photon-assisted tunneling of Cooper pairs through the junction (5). Because the height of the steps depends only on the frequency of the microwave radiation, Planck's constant, and the electric charge, Shapiro steps are nowadays used as the voltage standard [see, e.g., (12)]. Shapiro steps in superfluid ³He have been investigated (13).

The Josephson effect can also be observed in ultracold quantum gases (14). This enables the fabrication of atomtronic circuits, which have developed into their own field of research in recent years (15–17). Additionally, the high degree of control and flexibility over ultracold quantum gases provides unprecedented access to the microscopic physics underlying the system. In particular, Josephson physics has been recently implemented and extensively studied in ultracold atomic systems, both theoretically (18–23) as well as experimentally in bosonic (14, 24–26) and fermionic (27, 28) systems. In implementations with ultracold atoms, the Josephson current-phase relation is realized by moving the barrier through an ultracold gas that is otherwise at rest (29, 30). Shapiro steps in ultracold gases are predicted

to appear if the linear translation of the barrier (corresponding to a dc particle current) is combined with a periodic modulation of the position of the barrier (corresponding to an ac particle current), thus emulating the external microwave radiation of the superconducting Josephson junction (31).

In this work, we demonstrate the emergence of Shapiro steps at an atomic Josephson junction and quantify their global and local properties. The steps occur in the chemical potential difference, in conjunction with a density imbalance across the junction. We confirm that the step heights in the chemical potential difference $\Delta\mu$ are quantized by the external driving frequency f_m —i.e., $\Delta\mu = n\hbar f_m$, where n is the step index and $\hbar$ is Planck's constant. By spatially resolving the atomic density, we study the microscopic dynamics and observe the propagation of phonons and density depletions emerging from the barrier, which we identify as solitonic excitations. This paves the way for studying the microscopic dissipative dynamics of Shapiro steps and other related Josephson transport effects. Furthermore, given that the magnitude of the step height relates the chemical potential difference to a tunable frequency and that the density imbalance across the barrier is directly measurable, our results suggest a protocol to measure the equation of state of strongly correlated superfluids.

Experimental system

The experimental setup and the Shapiro protocol are sketched in Fig. 1. We prepare a ⁸⁷Rb Bose-Einstein condensate in an elongated optical dipole trap. The condensate has a tubelike cylindrical geometry with harmonic transverse confinement (Fig. 1A). The chemical potential is $\mu = \hbar \times 1900 \text{ s}^{-1}$. A movable repulsive barrier with height $V_B = 0.45 \times \mu$ separates the condensate into two parts and thus realizes a weak link (32). We probe the system by absorption imaging. To achieve a high accuracy, we perform a matter wave imaging scheme (33) to reduce the optical density (32).

We implement the dc and ac driving protocol following the proposal of Singh *et al.* (31). To this end, we move the barrier with constant velocity v and an additional periodic modulation

$$x(t) = vt + x_m \sin(2\pi f_m t) \quad (1)$$

where x_m is the modulation amplitude and f_m the modulation frequency. We fix the driving time to 33 ms, which means that the final position of the barrier depends on v . At the end, we measure the atom number imbalance $z = (N_R - N_L)/N$, where N is the total atom number and N_L and N_R are the atom numbers on the left- and right-hand sides, respectively, of the barrier. As a reference z_{ref} , we also measure the corresponding imbalance without the barrier. Using a simulation of the Gross-Pitaevskii equation, we map $\Delta z = z - z_{\text{ref}}$ to the chemical potential difference $\Delta\mu$ (32). The barrier motion induces an atomic current relative to the barrier of value I , with $I = vI_c/v_c$, where I_c is the critical current and v_c is the critical velocity (32). The external current then reads $I_{\text{ext}} = I + I_{\text{ac}} = I + I_m \cos(2\pi f_m t)$, with the bias current I and the modulation current I_m . The parameters that we vary in our protocol are I , I_m , and f_m . In Fig. 1C, we show the measurements of the current-chemical potential characteristics that are obtained without and with ac driving. From the undriven response, we characterize the critical velocity and the critical current of the Josephson junction (32). For the periodically driven case, the system features plateaus of constant chemical potential difference occurring at integer multiple of the driving frequency—the steps occur at lower I for stronger I_m . We model our system with extensive classical-field simulations (32), which capture the features of our driven Josephson junction well (Fig. 1, D and E).

In a superconducting Josephson junction, where the Shapiro steps arise from photon-assisted tunneling of the Cooper pairs (34), the voltage V_n of the n -th Shapiro step is directly given by the energy quantization—i.e., $V_n = n\Phi_0 f_m = n f_m \hbar / (2e)$, where Φ_0 is the flux quantum

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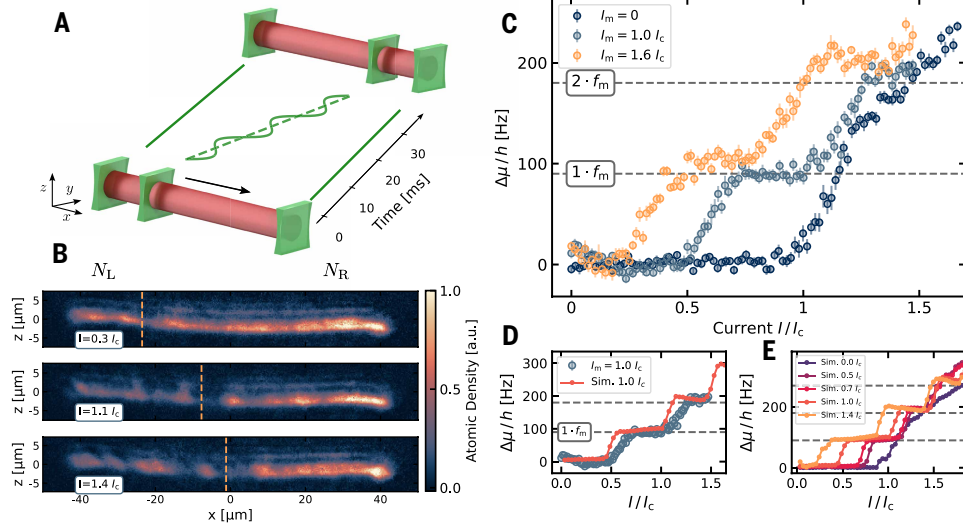


Fig. 1. Shapiro steps in an ultracold atomic Josephson junction. (A) The system consists of a cylindrically shaped superfluid, which is split by a movable optical barrier creating a weak link. The barrier as well as the two confining endcaps are realized with tightly focused laser beams. In the measurement protocol, the barrier is moved linearly (dc particle current) through the superfluid with an additional harmonic modulation (ac particle current), which is sketched in the figure. (B) At the end of the protocol, a real-space absorption image of the atomic cloud is taken, using matter wave imaging (32, 33). From the atom number imbalance between the two superfluids, their chemical potential difference is derived. The pictures are taken with a modulation current $I_m = 0.8I_c$. The position of the barrier is marked as a dashed orange line. The atomic density is given in arbitrary units (a.u.). (C) Measured Shapiro steps for $f_m = 90$ Hz and different modulation amplitudes I_m . The horizontal lines indicate the ideal Shapiro step height, given by Eq. 2. Every data point is the average of about 25 measurement runs, and the vertical bars indicate the errors of the mean. (D) Comparison between experiment and classical-field simulation at $T = 35$ nK (32). (E) Simulations of the system at different I_m (32).

and e the elementary charge (5, 35). For our neutral atom implementation, this results in the quantization condition

$$\Delta\mu = nhf_m \quad (2)$$

To confirm this fundamental relation, we vary f_m and determine the height $\Delta\mu$ of the first step (Fig. 2A). In Fig. 2B, we show the results for the simulation and for the experiment by averaging over several realizations with different modulation amplitudes. The simulation reproduces the relation trend accurately, but the experiment measures a 10% larger slope. The reason for this small deviation lies in the uncertainties of the experimental parameters (atom number and trapping potential), which enter the calculation of μ [see also (32)]. Taking this into account, we conclude that we indeed confirm the behavior $\Delta\mu = hf_m$ (Fig. 2B). Our results show that the driving protocol is capable of producing a predetermined chemical potential difference, which may enable the transfer of the voltage standard to the realm of ultracold quantum gases (23, 36). In addition, our results are useful in situations where the relation between the atomic density and the chemical potential, $\mu(n)$, is not known. Measuring the density difference at both sides of the junction directly reveals $d\mu(n)/dn$, which is referred to as the inverse thermodynamic density of states. Performing stepwise measurements by decreasing the density allows for the determination of the equation of state $\mu(n)$.

Next, we analyze the width of the Shapiro steps by varying the modulation amplitude I_m/I_c at a fixed modulation frequency of $f_m = 90$ Hz (32). In Fig. 2, C and D, the widths of the zeroth and first steps display a Bessel-function behavior as a function of I_m/I_c , which is in agreement with the results of numerical simulations (32). To support this observation, we use the analytical prediction of an ac voltage-driven Josephson junction, which yields the step width $I_n = I_c |J_n(V_m/V_n)|$, where J_n is the Bessel function of the n -th order and V_m the modulation voltage (5, 37). For an ac current-driven junction, this can be mathematically transformed into

$$I_n = I_c \left| J_n \left(\alpha \frac{I_m}{I_c} \right) \right| \quad (3)$$

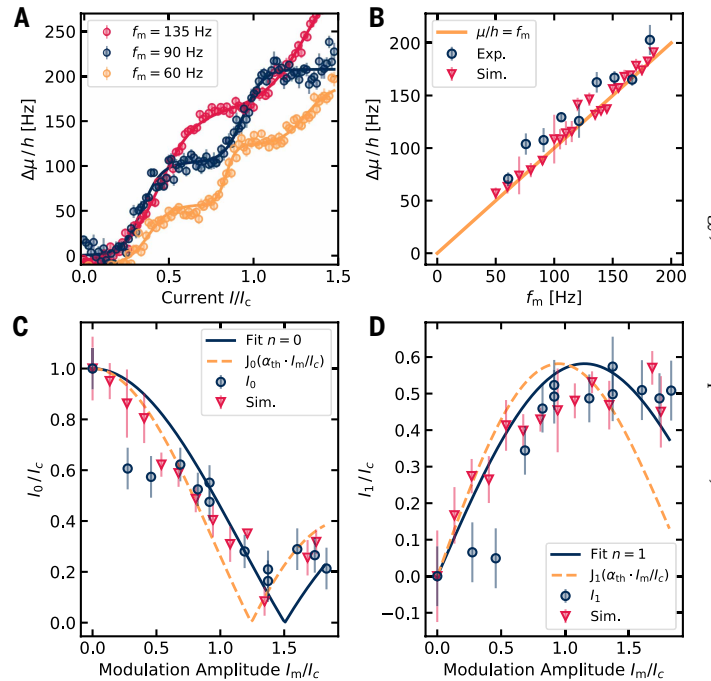


Fig. 2. Characteristics of the Shapiro steps. (A) Dependence of the step height on the modulation frequency f_m , where the step height is determined using sigmoid fits (continuous lines). (B) Measurements of the step height (dots) are compared with the numerical simulations (triangles) and the prediction $\Delta\mu/h = f_m$ (continuous line). (C and D) Measurements (dots) and simulation (triangles) of zeroth (C) and first (D) step widths (I_0 and I_1) for varying I_m/I_c . The solid blue lines are a fit to the experimental data, following Eq. 3, where both curves are fitted simultaneously with the same fit parameter α_{fit} , yielding $\alpha_{\text{fit}} = 2.15 \pm 0.08$. The dashed orange lines indicate the theoretical prediction, following Eq. 3 (see text).

with $\alpha = RI_c/f_m$. We use the independently determined system parameters R and I_c (32) to calculate this parameter, obtaining $\alpha_{th} = 1.93 \pm 0.07$. Using α_{th} , we plot the result of Eq. 3 as the orange dashed line in Fig. 2, C and D. Notably, this fit parameter free model supports the overall trend of our measurements and simulations.

Microscopic dynamics

Compared with solid-state physics, the time and length scales in quantum gas experiments are orders of magnitude larger, which enables the in situ study of collective excitations, quasiparticles, and related microscopic phenomena. To that end, we follow the time evolution of the real-space atomic density. Using only a constant barrier velocity—i.e., no ac particle current—we measure the dc and ac Josephson regime (Fig. 3A). Below the critical current, no imbalance is created, and only a single phononic density wave is emitted from the initial acceleration of the barrier (Fig. 3A, I and II). The group velocity of the emitted phonons coincides with a reference measurement, where a single perturbation creates a phonon wave packet (32). In the ac Josephson regime above the critical current, atoms are pushed away by the barrier, and a density depletion remains in its wake (Fig. 3A, III).

In the Shapiro regime, we see that the phase dynamics is instead characterized by two distinct collective excitations: phonons and localized density depletions (Fig. 3, B and C). Phonons are emitted at distinct times in both directions thanks to the oscillatory motion of the barrier. This is particularly visible on the zeroth plateau (Fig. 3, B and C, I). Increasing the constant current I beyond the zeroth plateau yields a chemical potential difference and atom number imbalance; this phenomenon is characterized by the formation of “depletion waves” that move backward compared with the barrier motion (blue lines propagating to the left in Fig. 3, B and C, II and III). These excitations are observed to have a smaller group velocity—i.e., a steeper slope (32). By additionally using the phase information of the simulation in Fig. 4A, we observe that such density depletions are accompanied by phase jumps across the density minimum and identify them as solitonic excitations (38). Analyzing the transverse phase profile right after the emission (32), we find that the solitonic excitations manifest themselves as vortex rings. These solitonic excitations are responsible for most of the reduction in density, ultimately causing the particle imbalance across the junction (Fig. 4, B and C). Specifically, the simulation shows that for the n -th plateau, n solitonic excitations are, on average, emitted backward during one driving cycle (Fig. 3C, II and III) (32). This creates the reduced density in the wake of the moving barrier and therefore gives rise to the resistive regime. This statement is supported by the fact that the rate of emitted solitonic excitations displays a steplike behavior in close parallel with the staircase displayed by the chemical potential (fig. S13). Depending on the system geometry, the emerging solitonic

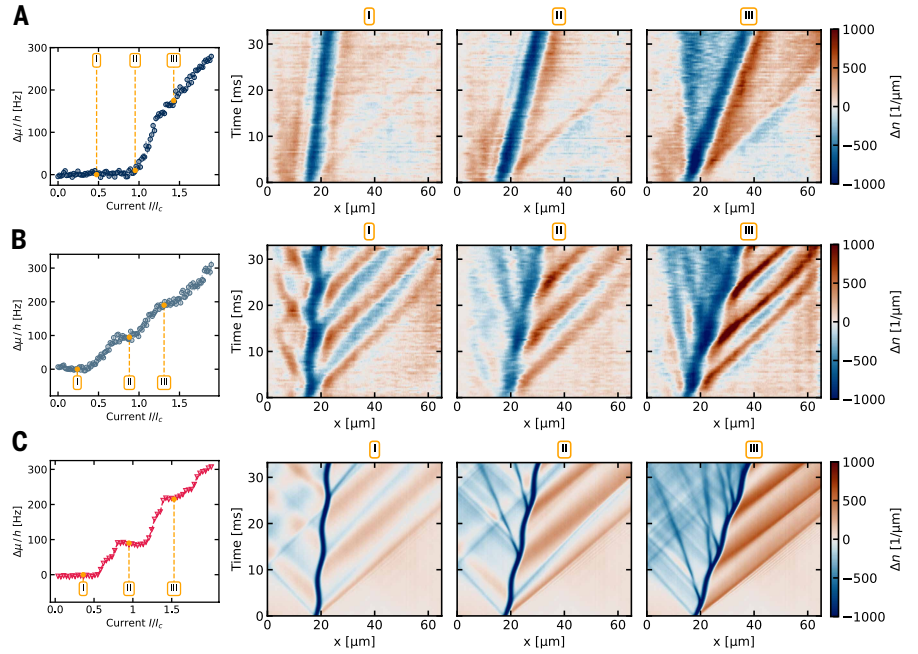


Fig. 3. Microscopic system dynamics. Time evolution of the atomic density in the experiment [(A) and (B)] and in the simulation (C). (A) Constant barrier velocity. We show the time evolution for three different currents, indicated in the $I - \Delta\mu$ plot on the left. (B) Constant plus periodically modulated barrier velocity (Shapiro protocol), where $I_m = 0.8I_c$ and $f_m = 90$ Hz. (C) Classical-field simulations of the driven junction, with $I_m = 0.8I_c$ and $f_m = 90$ Hz. Each horizontal line in the images corresponds to a transversely integrated absorption image, which are stacked together. The color code indicates the relative change of the line density with respect to a reference image without barrier.

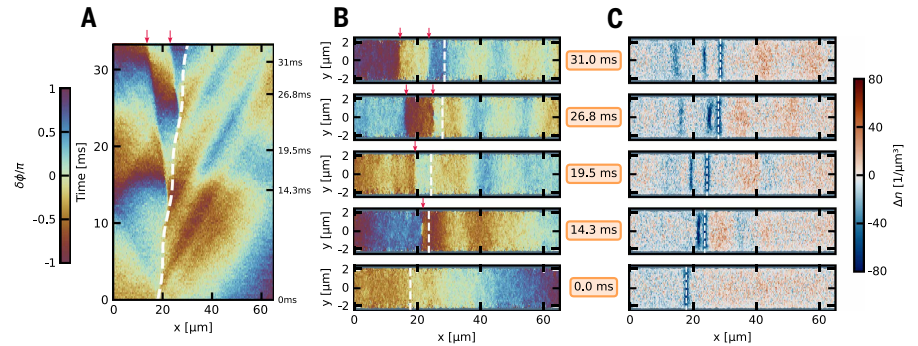


Fig. 4. Dynamics of solitonic excitations. (A) Time evolution of the phase $\delta\phi(x, t) = \phi(x, t) - \phi_m(t)$ along the central x line of a single trajectory of the system for $I/I_c = 0.9$, $I_m/I_c = 0.8$, and $f_m = 90$ Hz, where $\phi_m(t)$ is obtained after averaging the wave function over x . The periodic barrier motion (white dashed line) modifies the local phase near the barrier, which results in the creation of solitonic excitations moving to the left. The red arrows in (A) and (B) are a guide to the eye for the position of the solitonic excitations. (B and C) Snapshots of the phase profile $\delta\phi(x, y)$ of a single trajectory (B) and the averaged density profile $\Delta n(x, y)$ for the central z plane (C) at times $t = 0, 14.3, 19.5, 26.8$, and 31 ms. During each driving cycle, the barrier motion (white vertical dashed line) emits one solitonic excitation, whose subsequent time evolution manifests itself as density dips in (C). The positions of the phase jumps in (A) are marked with red arrows. At time 14.3 ms, the phase profile of a vortex ring is visible (32). There are also phononic excitations that propagate in both forward and backward directions. In (C), we show the relative change of the density with respect to a reference cloud without barrier. In (B) and (C), the thermal wings outside the Thomas-Fermi region of the condensate are shaded in gray.

excitations can be different. For instance, in the theoretical proposal (31), a larger two-dimensional system was used, and vortex-antivortex pairs were generated at the barrier. The appearance of vortex rings has been observed in previous studies (39, 40) as well as the emission of solitons from a barrier (41, 42).

We note that Shapiro steps have also been observed in a complementary experiment with ultracold fermions in the unitary regime and in a molecular Bose-Einstein condensate (43). Whereas in our work, vortex rings are the relevant solitonic excitations in the microscopic dynamics, the different geometry observed by Del Pace *et al.* (43) leads to the formation of vortex-antivortex pairs.

It is also instructive to compare the microscopic physical mechanism with that of superconductors. Our results effectively demonstrate that both superconducting and atomic junctions are characterized by coherent dynamics leading to a synchronization of the phase difference with the external drive (2). Whereas in superconductors, the tunneling of Cooper pairs in the voltage state is mediated by photons from the coherent driving field, the atomic superfluid couples to phonons when crossing the barrier at finite chemical potential difference.

As for the resistive channel, superconductors are characterized by an incoherent tunneling of quasiparticles (namely broken Cooper pairs), which are present in the superconductor at finite temperature (5). By contrast, in atomic superfluids, collective excitations, such as solitons and vortices, are the dominant dissipation mechanism, opening the resistive channel. Because they can be observed directly in the experiment, their role for the microscopic physics of Shapiro steps can be studied under clean conditions.

In superconductors, collective excitations have been investigated in complementary studies on Josephson physics, in which phase slips (44, 45), Josephson vortices (46), and solitons (47) have been observed experimentally. All of these observations have been made with static or pinned excitations; direct observation of the dynamics of such excitations remains challenging. However, we expect that collective phase excitations might also play a role in the Shapiro physics of superconductors, especially in regimes, for example, at very low temperatures, where the quasiparticle density is small.

Discussion and outlook

We have experimentally demonstrated the emergence of Shapiro steps in an ultracold quantum gas by emulating the external microwave field and an alternating particle current across the barrier with a periodic modulation of the barrier position. Using cold atom techniques, we could monitor the coherent dynamics of the system with unprecedented accuracy.

Extensions of the current work include different dimensions, geometries, and particle statistics, as well as superfluid mixtures, dipolar superfluids, and superfluids in optical lattices. These can generate other types of solitons, such as bright solitons or vortices, or defects related to an underlying lattice or competing orders. Equation 2 provides the possibility to measure the equation of state of strongly correlated many-body systems. In the context of superconducting Josephson junctions, our work motivates expanding the understanding of the microscopic dynamics of the Shapiro effect, including the role of collective excitations, such as solitons and vortices in regimes where the quasiparticle density is low. Furthermore, our results demonstrate the potential of atomic superfluids to act as quantum simulators for superconducting circuits.

Toward the development of atomtronic technology, Shapiro steps can be used to fine-tune quantum transport because they combine both superfluid and resistive transport. Stacks of Shapiro steps can be used to create even larger differences in chemical potential over a whole system or as a source for predetermined chemical potential differences.

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SUPPLEMENTARY MATERIALS

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Editor's summary

In the Josephson effect, a current can flow through an insulating barrier between two superconductors without dissipation. For a high enough current, dissipation does occur, and if microwave radiation is also applied, a series of plateaus—the so-called Shapiro steps—appear in the current-voltage characteristic. Instead of using superconductors, Del Pace *et al.* and Bernhart *et al.* observed Shapiro steps in an atomic Josephson junction. The researchers moved a barrier through an ultracold gas at a speed that had a periodically modulated component to emulate microwave radiation. The height of the resulting Shapiro steps in the chemical potential difference was proportional to the frequency of the modulation. —Jelena Stajic

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